

TS – Express

REST-Interface for Invoice

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1. Introduction

The TS-Express REST Interface for Invoice is designed to provide a standardized and efficient way to exchange invoice data between external systems and the TS-Express platform. This interface enables seamless integration of invoice-related processes such as data retrieval, validation, creation, and updates using HTTP-based communication and JSON data structures.

The main purpose of this interface is to simplify the interaction with the invoice module of the TS-Express system, reduce manual processing efforts, and ensure data consistency across systems. By using clearly defined endpoints, third-party applications can automate invoice workflows, check for errors before submission, and maintain up-to-date billing records in real time.

This documentation outlines the structure and usage of each available endpoint, along with the expected request and response formats, in order to support developers in implementing reliable and secure integrations.

2. History

2025-08-14 Version 2.02

the keys in JSON are defined in camelCase.
subPosition -> subPositions as JSON array
taxTotals as JSON array
freeAddress as JSON array
invoiceInfo added

2025-08-20 Version 2.03

JSON Invoice Defintion:

- renaming of head to header
- key "mainNr" and "mainNr2" moved to object "transportData"
- "creator" and "sentBy" object removed
- added the new key "creator" with one value, creator of the invoice document
- added the new key "exchangeRateDate"
- added the key "match" in the object "customer", "client" and "creatorAddress"

update the "Search" command

2025-08-28 Version 2.04

JSON Invoice Defintion Search "searchOption":

- renaming of date to invoiceDate
- renaming of to to toDate
- renaming of from to fromDate
- change invoiceDate of JSON_ARRAY to JSON_OBJECT

JSON Invoice Defintion Invoice:

Add "invoiceID"

Changed Address JSON

Defined Mandatory Fields

Added Comments for the JSON-Records

Added Section for Testenvironment

2025-09-10 Version 2.05

JSON Invoice Defintion Add "workflowData":

Add "wfId"

Add "wfNext"

Add "wfStatus"

Change on "4.Test Environment":
ADD "VALIDATE"
Change Error Response on "CREATE"

2025-09-19 Version 2.06

5.6 Error Handling and 5.3 Validation
Changes:
merged ErrorJSON with invoiceHomer Validation JSON
Return successful ValidationJSON

3. Configuring the interface

To use the TS Invoice REST Interface you first of all need to either install the http-task on your domino-server, or you ask your TS Express Provider to let you know the public URL of your service, if available.

3.1. Installation of the http-task on your domino-server

1. Open the Domino Console

Access the server console via the Domino Administrator client or remotely via terminal if enabled.

2. Start the HTTP Task

Type the following command into the server console:

load http

```
load http
```

3. Check via Task List (optional)

To ensure the task is active, you can list all running tasks using:

Show tasks

```
show tasks
```

4. Automatic Startup (optional)

To start the HTTP task automatically when the server boots, add the following line to the notes.ini file:

```
ServerTasks=...,HTTP
```

Make sure that port 80 (or 443 for HTTPS) is open and not blocked by a firewall. Also ensure that the server's Internet Site documents or Web configurations are correctly set up to respond to requests.

In case of problems, please contact the TransSoft Support.

3.2. Local URL

If you are running the REST-Service on your local Domino-Server, you will access the REST-Service with a local URL of your network.

The URL will be

<<IP-Adress domino-server>>/<<Path-Of-Portal-Database>>/REST.xps/Invoice/<<Command>>

Here is an example:



3.3. Public URL

If you are using the REST-Service from TransSoft directly, TransSoft will assign you a special URL for public access. Authentication will be based on username/password and maybe a special token.

3.4. HTTP-Service and URL

To interact with the interface, a HTTP-POST service must be implemented. Both the request and the response are formatted as JSON objects. Depending on the desired operation (SEARCH, GET, VALIDATE, CREATE, UPDATE), different endpoints are used. These endpoints must be adapted to the actual database path in your environment.

SEARCH	<<IP>>/<Path-Of-Portal-Database>>/REST.xps/Invoice/SEARCH
GET	<<IP>>/<Path-Of-Portal-Database>>/REST.xps/Invoice/GET
VALIDATE	<<IP>>/<Path-Of-Portal-Database>>/REST.xps/Invoice/VALIDATE
CREATE	<<IP>>/<Path-Of-Portal-Database>>/REST.xps/Invoice/CREATE
UPDATE	<<IP>>/<Path-Of-Portal-Database>>/REST.xps/Invoice/UPDATE

The URL is case-sensitive !

4. Test Environment

For testing of your system, you can use the TransSoft-Testenvironment. For this you need to have a login, which is assigned to you by the support of transsoft. Please get in contact with tssupport@transsoft.de.

The following test-URLs are available:

["https://transsoft.de/web/InvoicePortal.nsf/REST.xsp/Invoice/SEARCH"](https://transsoft.de/web/InvoicePortal.nsf/REST.xsp/Invoice/SEARCH)
["https://transsoft.de/web/InvoicePortal.nsf/REST.xsp/Invoice/CREATE"](https://transsoft.de/web/InvoicePortal.nsf/REST.xsp/Invoice/CREATE)
["https://transsoft.de/web/InvoicePortal.nsf/REST.xsp/Invoice/VALIDATE"](https://transsoft.de/web/InvoicePortal.nsf/REST.xsp/Invoice/VALIDATE)
["https://transsoft.de/web/InvoicePortal.nsf/REST.xsp/Invoice/GET"](https://transsoft.de/web/InvoicePortal.nsf/REST.xsp/Invoice/GET)

The best way to make an initial test is to create an invoice and read the invoice again with a get. Further you can try the search with some examples.

Comment: The Testenvironment is NOT protected against each others test-users. All Testusers are running under the same access of Agent "9990". Therefore be careful to NOT upload real invoices to the system.

SEARCH Example By Date:

<https://transsoft.de/web/InvoicePortal.nsf/REST.xsp/Invoice/SEARCH>

POST_JSON:

```
{
  "header": {
    "agent": "9990",
    "Type": null
  },
  "searchOption": {
    "adresscode": null,
    "adressMatch": null,
    "costCenter": null,
    "foreignkey": null,
    "invoiceDate": {
      "fromDate": "20250701",
      "toDate": "20250830"
    },
    "invoiceID": null,
    "invoiceNumber": null
  }
}
```

Result JSON:

```
{"searchResult":[{"invoiceId":"91BF609B656F06F0C1258CE5006295EC"},{"invoiceId":"6F265F4822EB729EC1258CC90053C80A"}]}
```

CREATE Example:

<https://transsoft.de/web/InvoicePortal.nsf/REST.xsp/Invoice/CREATE>

POST_JSON:

```
{“HomerInvoice”:.”...”}
```

Result JSON:

```
{“invoiceId”:“AD9B09613529EE3AC1258CF400577FBC”,“invoiceNumber”:“9990.000.000049”}
```

If creation was unsuccessful, an ErrJSON is returned. See “5.6 Error Handling.”

VALIDATE Example:

<https://transsoft.de/web/InvoicePortal.nsf/REST.xsp/Invoice/VALIDATE>

POST_JSON:

```
{“HomerInvoice”:.”...”}
```

Result JSON:

```
{“HomerInvoice”:.”...”}
```

In case of error, the ErrJSON is returned. See “5.6 ErrorHandling.” Otherwise, the same homerInvoice{} JSON object is returned. The correctly transmitted fields are empty in the returned HomerInvoiceJSON.

GET Example:

<https://transsoft.de/web/InvoicePortal.nsf/REST.xsp/Invoice/GET>

POST_JSON:

```
{
  "header": {
    "date": "Format: YYYYMMDD",
    "time": "Format: HHMMSS",
    "type": "GET",
    "agent": "9990",
    "version": "2.2.2"
  },
  "invoiceId": "AD9B09613529EE3AC1258CF400577FBC",
  "invoiceNumber": "",
  "foreignKey": ""
}
```

Result JSON:

```
{
  "header": {
    "date": "20250828",
    "time": "180057",
    "type": "INVOICE",
    "agent": "9990",
    "messageOption": "9",
    "version": "2.2.2",
    "invoice": {
      "invoiceData": {
        "invoiceType": "IO",
        "invoiceInfo": {
          "invoiceNumber": "9990.0000.000049",
          "invoiceNumberOrigin": "",
          "foreignKey": "",
          "costCenter": "",
          "creditorAccount": "",
          "debtorAccount": "10010",
          "performDate": "20250828",
          "invoiceDate": "20250821",
          "bookingDate": "",
          "netAmount": 3350.0,
          "taxAmount": 589.0,
          "amount": 3939.0,
          "currency": "EUR",
          "exchangeRate": 1.0,
          "exchangeCurrency": "EUR",
          "exchangeAmount": 3939.0,
          "exchangeRateDate": "",
          "payConditions": "",
          "creator": "Reiner Guenther/DE/Transsoft",
          "transportData": {
            "forwardingNr": "9991.0000F000016",
            "shipment": "",
            "sender": "adidas Sourcing Limited Taikoo Shing, Island East HK",
            "receiver": "Adidas Spor Malzemeleri Satış Ve Pazarlama 34387 Şişli, İstanbul TR",
            "pickupCity": "Petalina"
          },
          "deliveryCity": "Istanbul",
          "incoterm": "DDP",
          "colli": 50.0,
          "weight": 100.0,
          "payableWeight": 0.0,
          "volume": 3.0,
          "ldm": 0.0,
          "regionFrom": "KUL",
          "regionTo": "IST",
          "value": 0.0,
          "distance": 0.0,
          "pallet": 0.0,
          "tid": "",
          "reference": "",
          "orderNumber": "",
          "mainNr": "",
          "mainNr2": "",
          "customer": {
            "customerNr": "9992.0000.000003",
            "controlKey": "",
            "foreignKey": "",
            "match": "Adidas - Hong Kong",
            "name1": "adidas Sourcing Limited",
            "name2": "",
            "street": "",
            "city": "adidas Sourcing Limited",
            "zip": "",
            "countryCode": "HK",
            "phone": "",
            "fax": "",
            "vatId": "",
            "taxId": "",
            "mail": "",
            "companyId": "",
            "currency": "",
            "creditorAccount": "",
            "debtorAccount": "",
            "client": {
              "customerNr": "9990",
              "controlKey": "9990",
              "foreignKey": "",
              "match": "Adidas - Hong Kong",
              "name1": "adidas Sourcing Limited",
              "name2": "",
              "street": "",
              "city": "adidas Sourcing Limited",
              "zip": "",
              "countryCode": "HK",
              "phone": "",
              "fax": "",
              "vatId": "",
              "taxId": "",
              "mail": "",
              "companyId": "",
              "currency": "",
              "creditorAccount": "",
              "debtorAccount": "",
              "creatorAddress": {
                "customerNr": "9990",
                "controlKey": "9990",
                "foreignKey": "",
                "match": "Universal Exports GmbH",
                "name1": "Universal Exports GmbH",
                "name2": "",
                "street": "Elbchaussee 32",
                "city": "Hamburg",
                "zip": "22765",
                "countryCode": "DE",
                "phone": "+49 40 236742 - 0",
                "fax": "+49 40 236742 - 29",
                "mail": "impexp@importexport.com",
                "vatId": "DE103323874",
                "taxId": "103 - 323874",
                "companyId": "",
                "currency": "",
                "creditorAccount": "",
                "debtorAccount": ""
              },
              "subPositions": [
                {
                  "costCode": "1000",
                  "accRev": "8203",
                  "description": "Freight",
                  "addDescription": [
                    {
                      "key": "EN",
                      "text": "Freight"
                    },
                    {
                      "key": "RU",
                      "text": "Fracht"
                    },
                    {
                      "key": "KK",
                      "text": "Freight"
                    }
                  ],
                  "unit": "Stk",
                  "quantity": 1.0,
                  "unitPrice": 3000.0,
                  "amount": 3000.0,
                  "tax": 0.19,
                  "taxCat": "M2",
                  "taxAmount": 570.0,
                  "totalAmount": 3570.0
                },
                {
                  "costCode": "1600",
                  "accRev": "8203",
                  "description": "Handling",
                  "addDescription": [
                    {
                      "key": "EN",
                      "text": "Handling"
                    },
                    {
                      "key": "RU",
                      "text": "Handling"
                    },
                    {
                      "key": "KK",
                      "text": "Handling"
                    }
                  ],
                  "unit": "Stk",
                  "quantity": 50.0,
                  "unitPrice": 2.0,
                  "amount": 100.0,
                  "tax": 0.19,
                  "taxCat": "M2",
                  "taxAmount": 19.0,
                  "totalAmount": 119.0
                },
                {
                  "costCode": "2020",
                  "accRev": "8203",
                  "description": "Documentation",
                  "addDescription": [
                    {
                      "key": "EN",
                      "text": "Documentation"
                    },
                    {
                      "key": "RU",
                      "text": "Dokumentation"
                    },
                    {
                      "key": "KK",
                      "text": "Documentation"
                    }
                  ],
                  "unit": "Stk",
                  "quantity": 1.0,
                  "unitPrice": 100.0,
                  "amount": 100.0,
                  "tax": 0.0,
                  "taxCat": "A0",
                  "taxAmount": 0.0,
                  "totalAmount": 100.0
                },
                {
                  "costCode": "3020",
                  "accRev": "8203",
                  "description": "Fuel Surcharge",
                  "addDescription": [
                    {
                      "key": "EN",
                      "text": "Fuel"
                    }
                  ]
                }
              ]
            }
          }
        }
      }
    }
  }
}
```

```
Surcharge"}, {"key": "RU", "text": "Kraftstoffzuschlag"}, {"key": "KK", "text": "Fuel  
Surcharge"}], "unit": "Stk", "quantity": 1.0, "unitPrice": 150.0, "amount": 150.0, "tax": 0.0, "taxCat": "A0", "taxAmount": 0.0, "totalAmount": 150.0}, {"taxTotals": [{"tax": "A0", "amount": 250.0, "taxAmount": 0.0, "totalAmount": 250.0, "exchangeAmount": 250.0}, {"tax": "M2",  
"amount": 3100.0, "taxAmount": 589.0, "totalAmount": 3689.0, "exchangeAmount": 3689.0}], "foreignInData": {"foreignInvoiceNu  
mber": "", "foreignInvoiceDate": "", "foreignInvoiceTaxTotal": 0.0, "foreignInvoiceTotal": 0.0, "foreignInvoiceTotalNet": 0.0}}
```

Python Example:

```
login = "User/Password"
login_bytes = login.encode('utf-8')
login64 = base64.b64encode(login_bytes)
slogin = login64.decode('utf-8')

header={'Authorization': 'Basic '+slogin, 'Content-type': 'plain/text; charset=utf-8'}
resturl="https://transsoft.de/web/InvoicePortal.nsf/REST.xsp/Invoice/GET"

payload = {
    "header": {
        "date": "", # YYYYMMDD
        "time": "", # HHMMSS
        "type": "GET",
        "agent": "9990",
        "version": "2.2.2"
    },
    "invoiceId": "AD9B09613529EE3AC1258CF400577FBC",
    "invoiceNumber": "",
    "foreignKey": ""
}

jsonstr = json.dumps(payload, indent=4)

result=requests.post(resturl,headers=header,data=jsonstr)
print(result.text)
```

5. Commands

The interface is accepting the following commands: SEARCH, GET, VALIDATE, CREATE, UPDATE

The following chapter describes the possible commands of the interface in detail.

5.1. SEARCH

The SEARCH-command searches for a specific invoice using different search-criterias. The criterias can be combined, which will combine the filters by an AND-Operation. The search-criterias are handed over by a JSON-Object.

To request search for invoices, you must send a POST request containing a JSON object

Your JSON needs to have following structure:

```
{
  "header": {
    "date": "Format: YYYYMMDD",
    "time": "Format: HHMMSS",
    "type": "SEARCH",
    "agent": "partner identification",
    "version": "2.2.2"
  },
  "searchOption": {
    "invoiceId": "",
    "invoiceNumber" : "",
    "foreignKey" : "",
    "costCenter" : "",
    "addressMatch" : "",
    "addressCode" : "",
    "invoiceDate" : { "fromDate": "", "toDate": "" },
  }
}
```

invoiceId (unique) is the internal Number (Primary Key) of an Invoice in the Invoice-System

invoiceNumber (unique) is the public/external Number of an Invoice in the Invoice-System

costCenter is the Positionnumber of Invoices, especially in forwarding business

foreignKey(unique) is the PrimaryKey of an Invoice in the local Application

addressMatch is the searchcode of an address.

addresscode is the unique address-number of an address
date is a datereange (from-to) of the invoice-(registration)-date. You can select from to or just from.

At least one of the keys must contain a valid value; the others can remain empty. If more than one criterias are sent, all of them are combined with an "AND"-Operation.

Comment: Other search-options will follow

The SEARCH-Command will return a list of Invoices as JSON in the following structure

```
{ "searchResult": [ { "invoiceId": "<<value>>" }, { "invoiceId": "<<value>>" },  
{ "invoiceId": "<<value>>" } ... ] }
```

Example:

POST_JSON:

```
{  
  "header": {  
    "agent": "9990",  
    "Type": null  
  },  
  "searchOption": {  
    "adresscode": null,  
    "adressMatch": null,  
    "costCenter": null,  
    "foreignkey": null,  
    "invoiceDate": {  
      "fromDate": "20250701",  
      "toDate": "20250830"  
    },  
    "invoiceID": null,  
    "invoiceNumber": null  
  }  
}
```

Result JSON:

```
{ "searchResult": [ { "invoiceId": "91BF609B656F06F0C1258CE5006295EC" }, { "invoiceId": "6F265F4822EB729EC1258CC90053C80A" } ] }
```

5.2. GET

The GET-Command allows you to retrieve an invoice based on specific unique parameters.

To request invoice data, you must send a POST request containing a JSON object. This object must include at least one identifying element such as invoiceNumber, positionNo or foreignKey.

The response will return the relevant invoice data matching the criteria provided.

Your JSON needs to have following structure:

```
{
  "header": {
    "date": "Format: YYYYMMDD",
    "time": "Format: HHMMSS",
    "type": "SEARCH",
    "agent": "partner identification",
    "version": "2.2.2"
  },
  "invoiceId": "",
  "invoiceNumber": "",
  "foreignKey": ""
}
```

invoiceId (unique) is the internal Number (Primary Key) of an Invoice in the Invoice-System

invoiceNumber (unique) is the public/external Number of an Invoice in the Invoice-System

foreignKey is the PrimaryKey of an Invoice in the local Application

Only one of the fields can be filled.

The result will be an exactly one Invoice as Invoice-JSON.

Example:

POST_JSON:

```
{
  "header": {
    "date": "Format: YYYYMMDD",
    "time": "Format: HHMMSS",
    "type": "GET",
    "agent": "9990",
    "version": "2.2.2"
  },
}
```



```
"invoiceId":"AD9B09613529EE3AC1258CF400577FBC",  
"invoiceNumber":"","  
"foreignKey" : ""  
}
```

Result JSON:

```
{  
  "header": {  
    "date": "20250828", "time": "180057", "type": "INVOICE", "agent": "9990", "messageOption": "9", "version": "2.2.2",  
    "invoice": {  
      "invoiceData": {  
        "invoiceType": "IO", "invoiceInfo": "", "invoiceNumber": "9990.0000.000049", "invoiceNumberOrigin": "",  
        "foreignKey": "", "costCenter": "", "creditorAccount": "", "debtorAccount": "10010", "performDate": "20250828",  
        "invoiceDate": "20250821", "bookingDate": "", "netAmount": 3350.0, "taxAmount": 589.0, "amount": 3939.0, "currency": "EUR",  
        "exchangeRate": 1.0, "exchangeCurrency": "EUR", "exchangeAmount": 3939.0, "exchangeRateDate": "", "payConditions": "",  
        "creator": "Reiner Guenther/DE/Transsoft", "transportData": {  
          "forwardingNr": "9991.0000F000016", "shipment": "", "sender": "adidas Sourcing Limited Taikoo Shing, Island East HK",  
          "receiver": "Adidas Spor Malzemeleri Satış Ve Pazarlama 34387 Şişli, İstanbul TR", "pickupCity": "Petaling Jaya",  
          "deliveryCity": "İstanbul", "incoterm": "DDP", "colli": 50.0, "weight": 100.0, "payableWeight": 0.0, "volume": 3.0, "ldm": 0.0,  
          "regionFrom": "KUL", "regionTo": "IST", "value": 0.0, "distance": 0.0, "pallet": 0.0, "tid": "", "reference": "", "orderNumber": "",  
          "mainNr": "", "customer": {  
            "customerNr": "9992.0000.000003", "controlKey": "", "foreignKey": "", "match": "Adidas - Hong Kong",  
            "name1": "adidas Sourcing Limited", "name2": "", "street": "", "city": "adidas Sourcing Limited", "zip": "",  
            "countryCode": "HK", "phone": "", "fax": "", "vatId": "", "taxId": "", "mail": "", "companyId": "", "currency": "",  
            "creditorAccount": "", "debtorAccount": "", "client": {  
              "customerNr": "", "controlKey": "", "foreignKey": "", "match": "Adidas - Hong Kong",  
              "name1": "adidas Sourcing Limited", "name2": "", "street": "", "city": "adidas Sourcing Limited", "zip": "",  
              "countryCode": "HK", "phone": "", "fax": "", "vatId": "", "taxId": "", "mail": "", "companyId": "", "currency": "",  
              "creditorAccount": "", "debtorAccount": "", "creatorAddress": {  
                "customerNr": "9990", "controlKey": "9990", "foreignKey": "", "match": "Universal Exports GmbH",  
                "name1": "Universal Exports GmbH", "name2": "", "street": "Elbchausse 32", "city": "Hamburg",  
                "zip": "22765", "countryCode": "DE", "phone": "+49 40 236742 - 0", "fax": "+49 40 236742 - 29",  
                "mail": "impexp@importexport.com", "vatId": "DE103323874", "taxId": "103 - 323874", "companyId": "",  
                "currency": "", "creditorAccount": "", "debtorAccount": "", "subPositions": [{  
                  "costCode": "1000", "accRev": "8203", "description": "Freight", "addDescription": [{  
                    "key": "EN", "text": "Freight"}, {  
                    "key": "RU", "text": "Fracht"}, {  
                    "key": "KK", "text": "Freight"}], "unit": "Stk", "quantity": 1.0, "unitPrice": 3000.0, "amount": 3000.0, "tax": 0.19,  
                    "taxCat": "M2", "taxAmount": 570.0, "totalAmount": 3570.0, "costCode": "1600", "accRev": "8203",  
                    "description": "Handling", "addDescription": [{  
                      "key": "EN", "text": "Handling"}, {  
                      "key": "RU", "text": "Handling"}, {  
                      "key": "KK", "text": "Handling"}], "unit": "Stk", "quantity": 50.0, "unitPrice": 2.0, "amount": 100.0, "tax": 0.19,  
                      "taxCat": "M2", "taxAmount": 19.0, "totalAmount": 119.0, "costCode": "2020", "accRev": "8203",  
                      "description": "Documentation", "addDescription": [{  
                        "key": "EN", "text": "Documentation"}, {  
                        "key": "RU", "text": "Dokumentation"}, {  
                        "key": "KK", "text": "Documentation"}], "unit": "Stk", "quantity": 1.0, "unitPrice": 100.0, "amount": 100.0, "tax": 0.0, "taxCat": "A0",  
                        "taxAmount": 0.0, "totalAmount": 100.0, "costCode": "3020", "accRev": "8203",  
                        "description": "Fuel Surcharge", "addDescription": [{  
                          "key": "EN", "text": "Fuel Surcharge"}, {  
                          "key": "RU", "text": "Kraftstoffzuschlag"}, {  
                          "key": "KK", "text": "Fuel Surcharge"}], "unit": "Stk", "quantity": 1.0, "unitPrice": 150.0, "amount": 150.0, "tax": 0.0, "taxCat": "A0",  
                          "taxAmount": 0.0, "totalAmount": 150.0, "taxTotals": [{  
                            "tax": "A0", "amount": 250.0, "taxAmount": 0.0, "totalAmount": 250.0, "exchangeAmount": 250.0, "tax": "M2",  
                            "amount": 3100.0, "taxAmount": 589.0, "totalAmount": 3689.0, "exchangeAmount": 3689.0}], "foreignInData": {  
                      "foreignInvoiceNumber": "", "foreignInvoiceDate": "", "foreignInvoiceTaxTotal": 0.0, "foreignInvoiceTotal": 0.0,  
                      "foreignInvoiceTotalNet": 0.0}}]  
            }  
          }  
        }  
      }  
    }  
  }  
}
```

5.3. VALIDATE

The VALIDATE-Command is used to validate invoice data before creation or update. The command will perform consistency checks and further checks (registration-formula, etc) as being done while creation or update, but no record will be generated.

You must send a `homerInvoice{}` JSON object containing the invoice data via POST. The system returns the same `homerInvoice{}` JSON object, including any validation errors. Errors are displayed in such a way that only the field values that cause an error are filled in and contain the value that the TSE invoice would expect. An incorrectly sent "0" (zero) would return the value "-99999". In addition, an error JSON data set may be included, see "5.6.Error handling."

If the `HomerInvoice` JSON is correct, you will receive the following JSON as a return:

```
{"validate":{"validStatus":"OK"}}
```

This allows you to verify that all required fields and formats are correct before proceeding with further operations.

5.4. CREATE

The CREATE-Command will create a new invoice in the database.

To create an invoice, send a POST request containing the invoice data in a `homerInvoice`-JSON-object. If successful, the system will respond with a JSON object that includes the newly assigned invoice number, confirming the invoice has been created.

```
{  
  "invoiceId": "E168690EB8C00821C1258CC400469F8A",  
  "invoiceNumber": "1234.0000.000001"  
}
```

If the invoice has not been created, you will receive an error JSON object. See “5.6 Error Handling.”

Example:

POST_JSON:

```
{“homerInvoice”: “...”}
```

Result JSON:

```
{“invoiceId”: “AD9B09613529EE3AC1258CF400577FBC”, “invoiceNumber”: “9990.0000.000049”}
```

5.5. UPDATE

The UPDATE-Command will update an existing invoice in the database, as long as update is still allowed.

To update an invoice, send a POST request with the full `homerInvoice`-JSON object. The interface will process the data and apply the necessary changes to the existing invoice record in the system.

If the update was ok, you will receive a JSON with the Invoice-ID and the Invoice-Number of the updated invoice.

```
{
  "invoiceId": "E168690EB8C00821C1258CC400469F8A",
  "invoiceNumber": "1234.0000.000001"
}
```

If the update was not possible because the invoice was already locked (registered, issued, downloaded), you will receive a JSON with the Invoice-ID and the NEW Invoice-Number of the newly created invoice.

Comment: THIS OPTION MAY BE PROHIBITED.

If the invoice was not created, you will receive a JSON object including any validation errors. Errors are shown in the way, that only the field-values, which produce an error, will be filled and will hold the value, which the TSE Invoice would expect. Wrongly send "0" (Zero) would return the value "-99999". All other values will be empty.

5.6. Errorhandling

If an Action was not possible, you will receive a JSON object including any validation errors.

Error-messages consist of a JSON including the error-code and the error-message in english language and in local language, if defined.

```
{
  "Errors": [
    {
      "errorCode": "INV-001",
      "errorMessage": "Invoice number is missing",
      "errorMessageLocal": "Rechnungsnummer fehlt"
    },
    {
      "errorCode": "INV-002",
      "errorMessage": "Invalid cost center",
      "errorMessageLocal": "Ungültige Kostenstelle"
    }
  ]
}
```

Further, if the error is coming from the consistency of the sent invoice, the wrong fields are shown in the way, that only the field-values, which produce an error, will be filled and will hold the value, which the TSE Invoice would expect. Wrongly send "0" (Zero) would return the value "-99999". All other values will be empty.

In this case, the entire validation invoice JSON is sent in combination with the error JSON:

```
{
  "Errors": [
    {
      "errorCode": "INV-001",
      "errorMessage": "Invoice number is missing",
      "errorMessageLocal": "Rechnungsnummer fehlt"
    },
    {
      "errorCode": "INV-002",
      "errorMessage": "Invalid cost center",
      "errorMessageLocal": "Ungültige Kostenstelle"
    }
  ],
  {"homerInvoice": "..."}
}
```

6. Handling of Addresses

Addresses are not transmitted explicitly through a separate request but are instead included implicitly as part of the invoice data transmission. When an invoice is received, the system automatically checks whether the referenced address already exists in the database. If the address is found, it will be updated if necessary to reflect the latest information. If it does not exist, the system will create a new address record. This process ensures that address information remains accurate and up to date without requiring additional interface calls.

You can download addresses only by submitting a get-request for an existing invoice.

7. HOMER-Invoice

7.1.HOMER-INVOICE XML (Common Record)

Homer - Invoice

05.02.2024

HOMER

+ HEAD

DATE
TIME
TYPE
MESSAGEOPTION
VERSION

Format: YYYYMMDD

Format: HHMMSS

INVOICE

1 = Deletion, 5 = Changed Info, 9 = Original

2.2.2

+ INVOICE

+ INVOICEDATA

INVOICETYPE
INVOICENUMBER

INVOICENUMBERORIGIN
FOREIGNKEY
MAINNR1
MAINNR2
COSTCENTER
CREDITORACCOUNT
DEBITORACCOUNT
PERFORMDATE
INVOICEDATE
BOOKINGDATE
NETAMOUNT
TAXAMOUNT
AMOUNT
CURRENCY
EXCHANGERATE
EXCHANGE CURRENCY
EXCHANGEAMOUNT
PAYCONDITIONS
SALESMANAGER

Type of Invoice (IO/II/CO/CI...)

Unique System-ID

Corresponding Unique System-ID to other Invoices

Customers Invoicekey

MainNr1 from Shipment / Transport Data Invoice

MainNr2 from Shipment / Transport Data Invoice

Voyagenumber / Positionnumber

Document is creditorial, if CredAccount is sent

Document is debitorial, if DebAccount is sent

YYYYMMDD

YYYYMMDD

YYYYMMDD

Total amount

Currency of the invoice

(Table - Payconditions)

Person created the Invoice

+ CREATOR

NAME
FIRSTNAME
PHONE
FAX
MAIL

+ SENTBY

NAME
FIRSTNAME
PHONE
FAX
MAIL

Seite 24

+	SUBPOSITION		
		COSTCODE	Costcode in the System
		ACCREV	Account Revenue (optional if available)
		DESCRIPTION	
	+	ADDDescription	optional
		KEY	language key
		TEXT	description in the language
	+	ADDDescription	optional
		KEY	language key
		TEXT	description in the language
		ADDDescription	optional
		KEY	language key
		TEXT	description in the language
		UNIT	Kind of unit
		QUANTITY	Count of units for the position
		UNITPRICE	Price per unit
		AMOUNT	Price for the position (Quantity*Unitprice)
		TAX	Taxvalue for the position
		TAXCAT	Taxcategorie for the position
		TAXAMOUNT	Amount of tax for the position (Amount*Tax)
		TOTALAMOUNT	Total amount for the position (Taxamount+Amount)
+	TAXTOTALS		
		TAX	
		AMOUNT	
		TAXAMOUNT	
		TOTALAMOUNT	
+	EXCHANGEAMOUNT		
+	FOREIGNINDATA		
		FOREIGNINVOICENUMBER	
		FOREIGNINVOICEDATE	
		FOREIGNINVOICETAXTOTAL	
		FOREIGNINVOICETOTAL	

7.2. HOMER-INVOICE JSON

The Homer-Invoice-JSON-Record is derived from the HOMER-XML-record. Therefore all detailed descriptions can be found in the corresponding section of the HOMER-XML-record

For better display, all mandatory fields are shown in red. Orange lines are mandatory in special circumstances.

```
{
  "homerInvoice": {
    "header": {
      "date": "Format: YYYYMMDD",
      "time": "Format: HHMMSS",
      "type": "INVOICE",
      "agent": "partner identification",
      "messageOption": "1 = Deletion, 5 = Changed Info, 9 = Original",
      "version": "2.2.2"
    },
    "invoice": {
      "workflowData": {
        "wfId": "workflow position",
        "wfNext": "next user",
        "wfStatus": "status"
      },
      "invoiceData": {
        "invoiceType": "Type of Invoice",
        "invoiceInfo": "invoice info",
        "invoiceNumber": "Unique TSE Invoice-ID",
        "invoiceId": "Unique System-ID",
        "invoiceNumberOrigin": "Corresponding Unique System-ID to other Invoices",
        "foreignKey": "Customers Invoicekey",
        "costCenter": "Voyagenumber / Positionnumber",
        "creditorAccount": "creditor account",
        "debitorAccount": "debitor account",
        "performDate": "YYYYMMDD",
        "invoiceDate": "YYYYMMDD",
        "bookingDate": "YYYYMMDD",
        "netAmount": "net amount",
        "taxAmount": "tax amount",
        "amount": "gross amount",
        "currency": "Currency of the invoice",
        "exchangeRate": "exchangeRate currency = 1 exchangeCurrency",
        "exchangeCurrency": "main currency",
        "exchangeAmount": "gross amount in main currency",
        "exchangeRateDate": "YYYYMMDD",
        "payConditions": "(Table - Payconditions)",
        "creator": "creator name"
      },
      "transportData": {
        "forwardingNr": "Shipmentkey",
        "shipment": "Description of cargo / Shipmentdescription",
        "sender": "shipper address",
        "receiver": "receiver address",
        "pickupCity": "pickup city",

```

```
"deliveryCity": "delivery city",
"incoterm": "international commercial terms",
"colli": "colli",
"weight": "weight",
"payableWeight": "payable weight",
"volume": "volume",
"ldm": "Loadmeters",
"regionFrom": "from region",
"regionTo": "to region",
"value": "ware value",
"distance": "distance in km",
"pallet": "Count of pallets",
"tid": "Container-ID / Truck-ID",
"reference": "shipment reference",
"orderNumber": "Ordernumber of customer",
"mainNr1": "MainNr1 from Shipment / Transport Data Invoice",
"mainNr2": "MainNr2 from Shipment / Transport Data Invoice"
},
"customer": {...Address-Record... },
"client": {...Address-Record...},
"creatorAddress": {...Address-Record...},
"bankAccount":[
  {
    "accountNumber": "Account from Branch",
    "iban": "IBAN from Branch",
    "bank": {
      "name": "Bank name",
      "id": "Bank ID",
      "bic": "Bank BIC"
    }
  },
  {
    ... Bank 2
  }
]
"subPositions": [
  {
    "costCode": "Costcode in the System",
    "accRev": "Account Revenue (optional if available)",
    "description": "description",
    "addDescription": [
      {
        "key": "language iso code",
        "text": "description in the language"
      },
      {
        "key": "language iso code",
        "text": "description in the language"
      }
    ],
    "unit": "Kind of unit",
    "quantity": "Count of units for the position",
    "unitPrice": "price per unit",
    "amount": "price for the position (quantity * unit price)",
    "tax": "tax value for the position",
    "taxCat": "tax category for the position",
    "taxAmount": "Amount of tax for the position (amount * tax)",
    "totalAmount": "Total amount for the position (tax amount + amount)"
  },
]
```

```
{
  "costCode": "Costcode2 in the System (second subposition)"
  ....
}
],
"taxTotals": [
  {
    "tax": "tax value (for example 0.05 for 5%)",
    "amount": "net amount of this tax category",
    "taxAmount": "tax amount of this tax category",
    "totalAmount": "gross amount of this tax category"
  },
  {
    "tax": "",
    "amount": "",
    "taxAmount": "",
    "totalAmount": ""
  }
],
"foreignInData": {
  "foreignInvoiceNumber": "invoice number in the foreign system",
  "foreignInvoiceDate": "YYYYMMDD",
  "foreignInvoiceTaxTotal": "total tax amount",
  "foreignInvoiceTotal": "total gross amount",
  "foreignInvoiceTotalNet": "total net amount"
}
}
}
```

Comments:

- The Header must always be sent to identify the so called “Agent”, which is the branchcode of the agent defined in the destination system (e.g. 8411, 8414, 2411)
- Message Option is not defined so far
- invoiceNumber is a “speaking” number/code, which normally is unique and is used in client-communication
- invoiceID is unique primary key of the address in the destination-system (MM2)
- invoiceNumber and/or invoiceID can be sent only in case of an update. For new invoice (create) the fields will be empty or ignored. If known, they identify an existing invoice-record in the destination-system. Both values will be known as the result of the create-method.
- InvoiceNumberOrigin is the reference to the origin invoice for depending invoices like corrections. If a correction is created, this field is mandatory.
- ForeignKey is the unique primary key of the invoice in the source-system
- The Transport-Data-Section is optional.
- ForwardingNumber is the reference to the shipment in the source-system. (STN)
- MainNr1 and MainNr2 are the Forwarding-Document Number like (HB/L, MB/L, CMR ...)
- Customer is the payer of the service. It is mandatory.
- Client is the buyer of the service. It can be different of the Customer.
- CreatorAddress is the Address-Record of the Creating Branch, like 8411, 8415. It should be corresponding to the agent-field in the header.
- Bankaccount is optional. It can define the account being printed on the invoice. Normaly it will be overruled by setting of the destination-system.
- Subpositions are the single invoice-lines. At least one line is mandatory.

- TaxTotals is an array of the summary of all net-amounts and it's calculated taxes. It is mandatory to verify the single values of the invoice. All values must be consistent.
- ForeignData is used for incoming Documents (II, CI) to transmit the foreign invoice-figures (Date, InvoiceNumber) and to verify the single values of the invoice. All values must be consistent.
- /homerInvoice/invoice/workflowData/wfStatus describes the status of the invoice;
"0" - means that the invoice has not yet been registered and the invoice can be changed.
"1" – the invoice is registered and the invoice may no longer be changed
"2" – the invoice is issued and the invoice may no longer be changed
"3" - The invoice has been transferred to the accounting department and the invoice may no longer be changed ("Downloaded")

7.3.Address-Records

For better display, all mandatory fields are shown in red. Orange lines are mandatory in special circumstances.

```
"address": {  
  "addressNumber": "Unique System-Addressnumber",  
  "addressId": "Unique System-ID",  
  "controlKey": "Unique Forwarder-ID",  
  "foreignKey": "Additional key",  
  "match": "match",  
  "name1": "name 1",  
  "name2": "name 2",  
  "department": "department",  
  "firstName": "firstname",  
  "middleName": "middlenames",  
  "lastName": "lastname",  
  "title": "title",  
  "street": "street",  
  "poBox": "postbox if available",  
  "city": city,  
  "zip": "postal city code",  
  "countryCode": "Defines the Country (ISO-Codes)",  
  "phone": phone number,  
  "fax": fax number,  
  "email": mail address,  
  "vatId": vat id,  
  "taxId": tax id,  
  "companyId": "e.g. OKPO-Code / Traderegisternumber",  
  "currency": "Standart Currencycode (ISO-Codes) for the customer",  
  "creditorAccount": creditor account,  
  "debitorAccount": debtor account,  
  "freeAddress":  
    "name street zip city country", for example, string, could contains line  
    breaks  
  "comment": "free comments and informations"  
}
```

Comments:

- „address“ is classtype, Addresses can be used as any kind of address-submission (e.g. Customer. Client, Sender etc)
- addressNumber is a “speaking” number/code, which normally is unique and is used in client-communication
- addressID is unique primary key of the address in the destination-system (MM2)
- adressNumber and/or addressID can be sent, if knowm. It identifies an existing address-record in the destination-system. It can be received by diownloading an existing invoice and check the address.id/address-Number in the client-section.

- If addressID and addressNumber are not send, the destination-system (MM2) will try to identify the address based on the sent fields. If no address can be found, it automatically will be generated.
- ControlKey is the code of special Agents, like 8411, 8415, 2411
- ForeignKey is the unique primary key of the address in the source-system
- Either Matchcode and Name1 must be sent (if address is a company) or lastname and title (if address is a real person). If both is sent, the person will be a contact person.
- FreeAddress is a string representing the address in any kind of special language, like Greek, Farsi, Chinese. **It can be in Unicode or must be coded as (not defined so far).** It also can be used for special kind of addressing e.g. with apartment or other kind of addressidentifikation.

8. Sequence

